



Anatomy of Flowering Plants



Key Tissues in Plants

- Meristematic tissue
- Permanent tissue
 - Simple
 - Complex

Anatomy is the study of internal structure of plants. It helps in understanding the functions performed by different tissues and organs. ♥

① Anatomy of Dicot Root

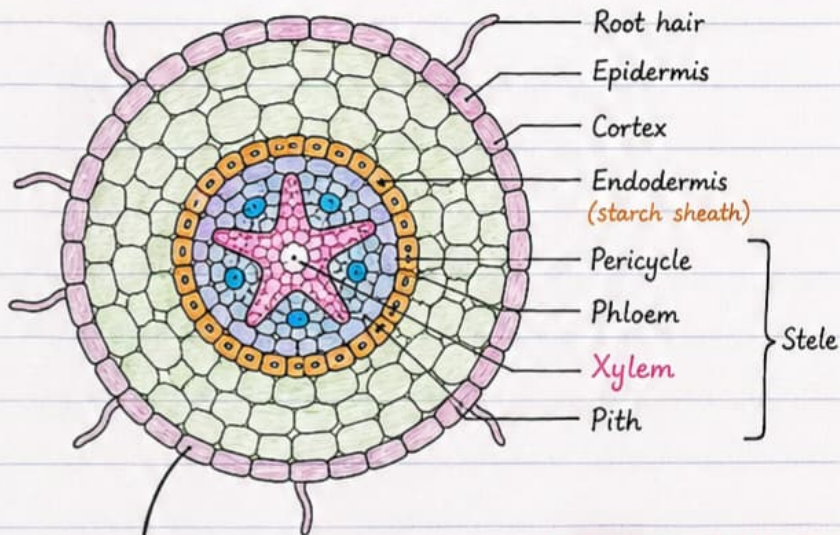
T.S. of Dicot Root

a) Epidermis

- Outermost layer of root.
- Made up of compactly arranged cells.
- Bears **root hairs** → increase surface area for absorption.

b) Cortex

- Below epidermis.
- Made up of parenchymatous cells.
- Stores food.
- Inner most layer → Endodermis (starch sheath) with **Casparian strip**.

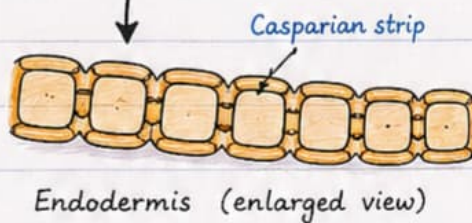


c) Stele

- Central part.
- Contains pericycle, vascular tissues and pith.

d) Pericycle

- Single layered cells.
- Helps in lateral root formation.



Casparian strip prevents back flow of water and ensures unidirectional movement.

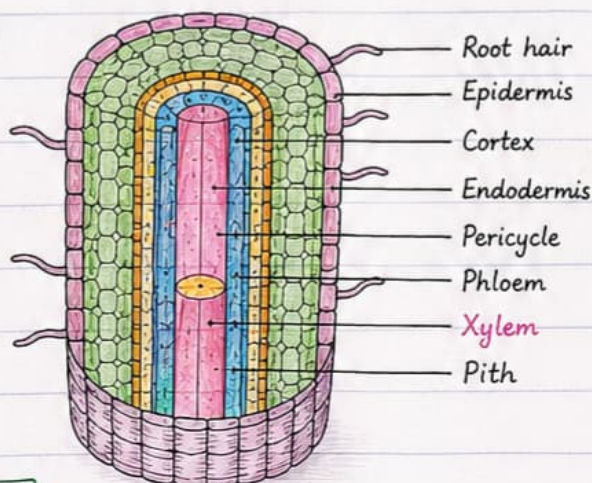
e) Vascular tissue

- Xylem & Phloem arranged in radial manner.
- Xylem is **star shaped** (exarch).
- Phloem lies between the arms of xylem.

f) Pith

- Small, parenchymatous cells at the centre.
- May be absent in some dicot roots.

3D View of Dicot Root



Functions

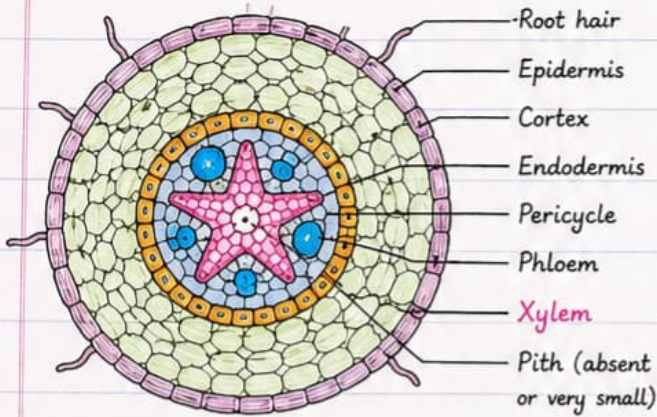
- Absorption of water & minerals
- Anchorage
- Storage
- Conduction
- Lateral root formation

Note In dicot root → xylem is exarch and phloem is endarch.

② Anatomy of Dicot & Monocot Root

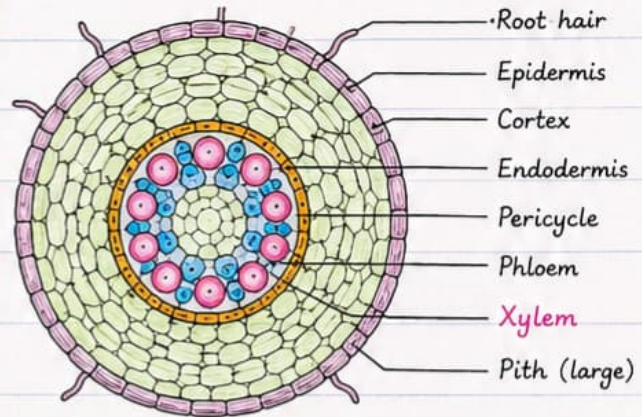


Dicot Root (T.S.)



- Xylem is star shaped (exarch).
- Pith is absent or very small.
- Number of xylem arms → 2 to 6.
- Common in dicot plants.

Monocot Root (T.S.)

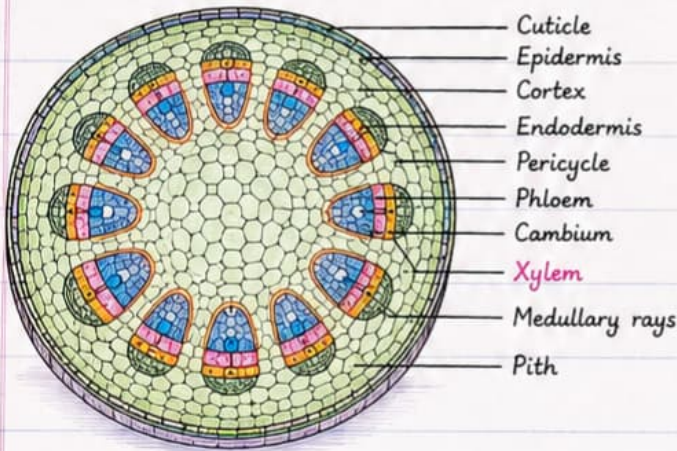


- Xylem and phloem are arranged in a ring (polyarch).
- Pith is large and well developed.
- Common in monocot plants.

③ Anatomy of Stem



(a) Dicot Stem (T.S.)



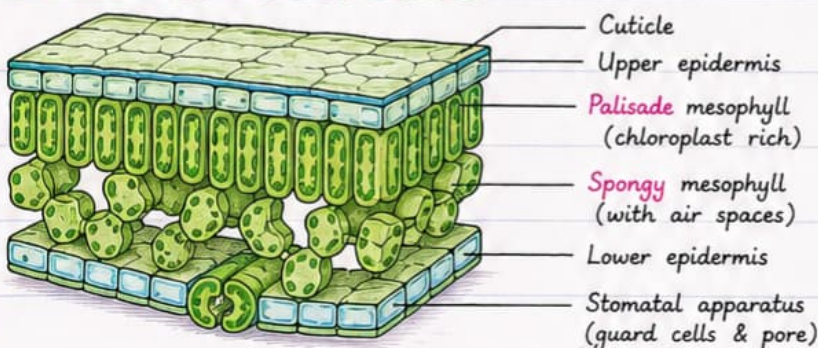
* Key Points *

- Vascular bundles are arranged in a ring.
- Conjoint, collateral and open vascular bundles.
- Cambium is present → allows secondary growth (increase in girth).
- Xylem lies towards inside and phloem towards outside.
- Pith is present and well developed.

Functions of Stem

- Transportation of water, minerals and food.
- Support to leaves, flowers and fruits.
- Storage of food (in some plants).
- Photosynthesis in green stems.

④ Anatomy of Leaf (T.S.)



Key points

- Leaves are **dorsiventral** in dicots.
- Palisade mesophyll → main site of photosynthesis.
- Spongy mesophyll → helps in gaseous exchange.
- Stomata → regulate exchange of gases and transpiration.

★ Key takeaway

Different tissues in roots, stems and leaves work together to perform important functions like absorption, conduction, support and photosynthesis.

Understand the structure, remember the function!



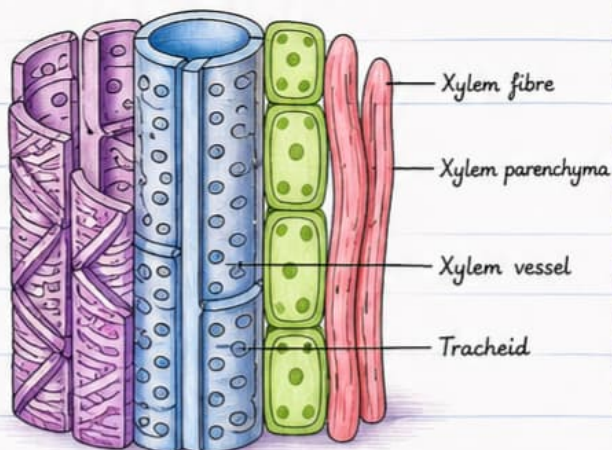
⑤ Vascular Tissues

Vascular tissues are complex tissues responsible for **transport**. ❤

(a) Xylem

3D View of Xylem

- Composed of tracheids, vessels, xylem parenchyma & xylem fibres.
- Xylem transports **water & minerals** from roots to other parts.
- It also provides **mechanical strength**.



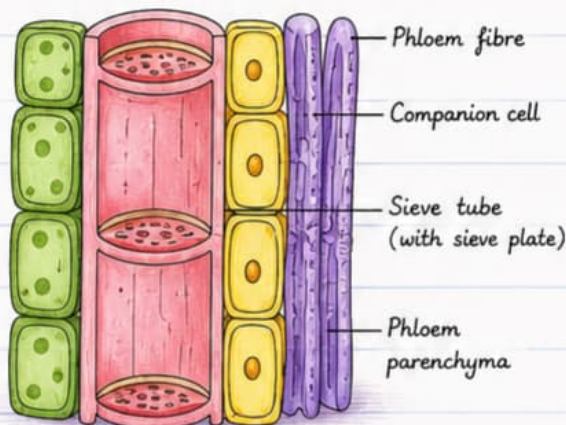
★ Key points - Xylem

- Tracheids → long, narrow with tapering ends.
- Vessels → wide cells, joined end to end.
- Fibres → provide mechanical strength.
- Parenchyma → storage of food.

(b) Phloem

3D View of Phloem

- Composed of sieve tubes, companion cells, phloem parenchyma & phloem fibres.
- Phloem transports **food** (mainly sucrose) from leaves to other parts (**translocation**).
- It conducts in both upward and downward directions.



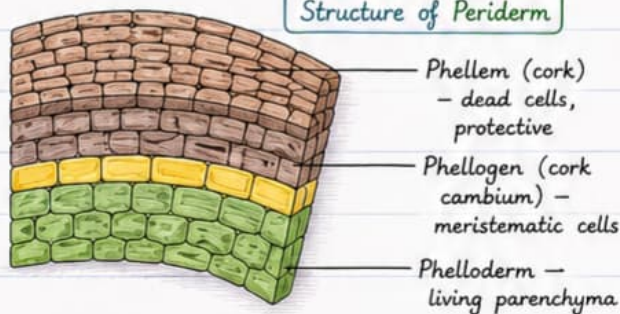
★ Key points - Phloem

- Sieve tubes → living cells, arranged end to end with sieve plates.
- Companion cells → help in conduction and provide energy.
- Parenchyma → storage of food.
- Fibres → mechanical strength.

⑥ Periderm (Protective Tissue in Older Roots & Stems)

- In older roots & stems, epidermis breaks and is replaced by **periderm**.
- Periderm is made up of three tissues - **phellem**, **phellogen** & **phelloderm**.
- It protects the plant from mechanical injury, water loss & infection.

Structure of Periderm



Functions

- Prevents water loss.
- Protects from mechanical damage.
- Protects from infection by pathogens.

⑦ Summary Table

Tissue System	Tissues Included	Location	Main Function
Dermal	Epidermis, Periderm	Outer covering of plant body	Protection, reduces water loss, prevents infection
Ground	Parenchyma, Collenchyma, Sclerenchyma	Cortex, Pith, Mesophyll, Xylem, Phloem	Support, storage, photosynthesis, packing
Vascular	Xylem, Phloem	In vascular bundles	Transport of water, minerals and food

Remember

Structure gives strength, function gives life. ❤



Understand the structure deeply, Nature will make sense beautifully!



⑧ Tissue Systems in Plants

Plant body is made up of three tissue systems. *

1. Dermal Tissue System

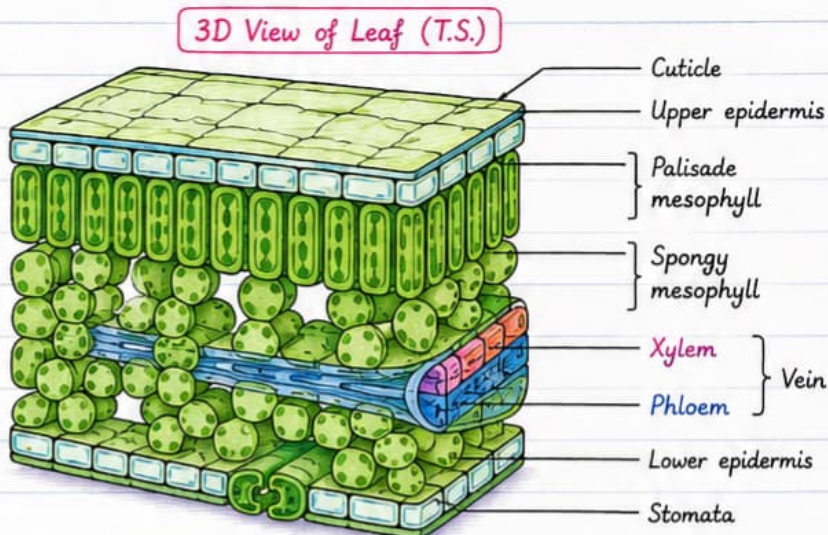
- Forms the outermost layer and protects the plant.

2. Ground Tissue System

- Forms the bulk of the plant body and performs functions like storage, photosynthesis and support.

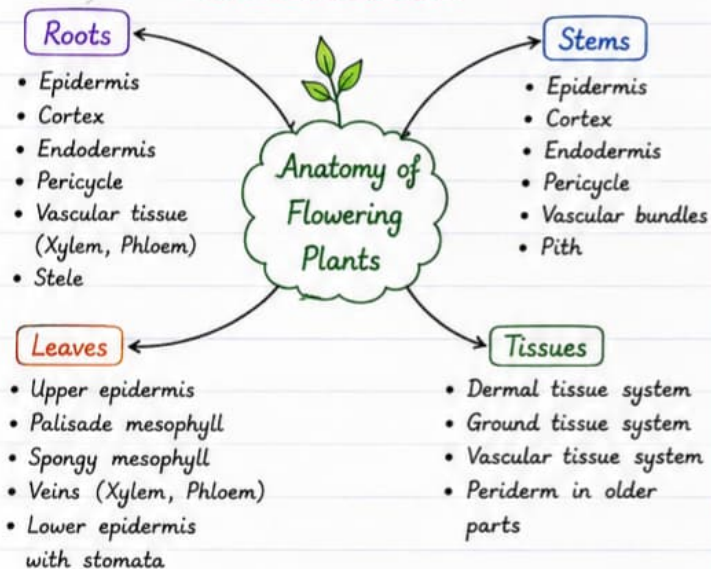
3. Vascular Tissue System

- Responsible for transport of water, minerals and food throughout the plant.



All tissues work together in a coordinated manner.

Quick Mind Map *



Important Points

- * Root hair increases surface area for absorption.
- * Endodermis has Casparian strip which controls the entry of water.
- * Xylem conducts water and minerals (mainly upward).
- * Phloem transports food (upward and downward).
- * Stomata help in gaseous exchange and transpiration.
- * Periderm replaces epidermis in old stems and roots.

Key Differences

Feature	Dicot Root	Monocot Root
Pith	Small or absent	Large and prominent
Xylem	Star shaped	Many (polyarch) in a ring
Number of xylem arms	2 to 6	Many (more than 6)
Example	Pea, Sunflower	Maize, Wheat

ONE SHOT REVISION

- ✓ Plants show well organized tissues.
- ✓ Three tissue systems – Dermal, Ground, Vascular.
- ✓ Root, stem and leaf have a specific arrangement of tissues.
- ✓ These tissues perform functions of absorption, transport, support, storage, protection and photosynthesis.
- ✓ Structure is related to function.

Understand the structure today, to remember the functions forever! ♥